

Luke Conibear

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Overview

Generalist engineer working across software engineering, data engineering, applied ML/data science, and atmospheric science to build production systems.

Experience

Senior Software Engineer, Tomorrow.io | 2022–present

- Primary contributor to a department-wide cloud cost optimisation program, delivering a 25% spend reduction in 2025 through information-content compression ([xbitinfo](#)), spot/reserved capacity strategy, KEDA scale-to-zero, rightsizing, lower cross-cloud/cross-region transfer, and stricter data-retention policies.
- Designed, deployed, and maintained 100+ production microservices and data pipelines in Python, Rust, and Go for ML and physics-based forecasting workflows ([YouTube](#)).
- Core contributor in Severity-1 incident response; identified recurring failure modes and led preventive fixes that reduced MTTR by 35% from 2024 to 2025.
- Led design and delivery of multi-cloud MLOps infrastructure across Azure ML, AWS SageMaker, and GCP Vertex AI.
- Led refactoring of legacy applications to improve operational reliability, maintainability, and simplicity.
- Led department-wide adoption of AI-assisted engineering workflows, integrating code review standards, clear documentation, and CI/CD improvements to increase delivery speed and code quality.
- Maintain event-driven distributed system (Go) that coordinates multi-cloud weather data pipelines (GCP, AWS, Azure) using PostgreSQL-backed dependency tracking to dispatch processing jobs when file inputs are met.

Research Software Engineer Machine Learning, University of Leeds | 2021–2022

- [Taught](#) AI emulators of numerical atmospheric models.
- Authored and maintained an open-source course for [High-Performance Python](#) and [Machine Learning](#).

Research Fellow, University of Leeds | 2018–2021

- Engineered [AI emulators](#) ([scikit-learn](#), [tpot](#)) to act as fast proxies for complex atmospheric simulations (~20 TB of data), distributing training across high-performance compute clusters ([dask](#)).
- Maintained and supported [WRFotron](#), an open-source automation tool for atmospheric modelling, managing GitHub releases, [documentation](#), and a community of users.

Education

- **PhD**, Ambient Air Quality and Human Health, University of Leeds | 2015–2018
– Computational modelling & applied data science
- **MSc**, Bioenergy, University of Leeds | 2014–2015
- **BEng**, Mechanical Engineering, University College London | 2007–2010

Publications

- Lead-author of [9 academic papers](#) (h-index of 18 with 1,400+ citations).

Tools

- **AI-native:** Cursor, Codex, GitHub Copilot.
- **Languages:** Python (scientific & ML focus), Rust, Go, Bash.
- **Cloud:** Azure (ML, AKS, storage, event grid), GCP (Vertex AI, GKE, GCS, Pub/Sub), AWS (SageMaker, S3).
- **Platform & Ops:** Git (GitHub), Kubernetes, Docker, Terraform, Argo, GitHub Actions, GPUs, DataDog, PagerDuty, DoIT, Jira.